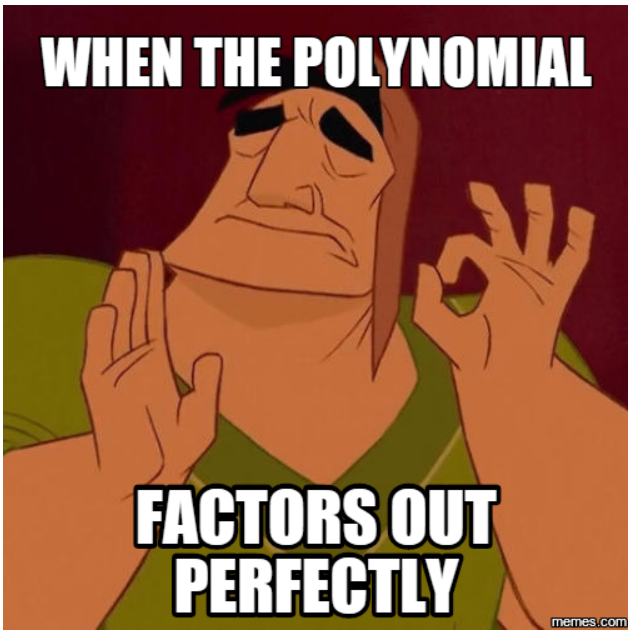

Grade 12 Advanced Functions
Polynomial Functions Key

First Name: _____**KEY**_____

Last Name: _____

Directions:

- Please answer to 2 decimal points
- The test is designed to be completed in 75 minutes

For grading use only

Page:	2	3	4	5	6	Total
Points:	8	10	14	18	10	60
Score:						

Multiple Choice (10 marks)

1. (1 point) The minimum number of turning points that a quartic function can have is:
   A. 0
   B. 1
   C. 2
   D. 3

2. (1 point) The degree of the polynomial $f(x) = -x^4 + 5x^3 - 6x^2 + 3$ is:
   A. 2
   B. 3
   C. 4
   D. 5

3. (1 point) The leading coefficient of $f(x) = -x^4 + 5x^3 - 6x^2 + 3$ is:
   A. -6
   B. 1
   C. 5
   D. -1

4. (1 point) Which of the following is NOT a factor of $x^3 - 5x^2 + 8x - 40$?
   A. $x - 5$
   B. $x + 4$
   C. $3x - 2$
   D. $2x - 3$

5. (1 point) The end behavior of $f(x) = -5x^4 + 3x^3 - 2x^2 + 6x - 9$ is:
   A. $x \to -\infty, f(x) \to -\infty$; $x \to \infty, f(x) \to -\infty$
   B. $x \to -\infty, f(x) \to \infty$; $x \to \infty, f(x) \to \infty$
   C. $x \to -\infty, f(x) \to \infty$; $x \to \infty, f(x) \to -\infty$
   D. $x \to -\infty, f(x) \to -\infty$**;** $x \to \infty, f(x) \to -\infty$

6. (1 point) When $f(x) = 3x^3 + 5x^2 - 2x + 8$ is divided by $x + 2$, the remainder is:
   A. 8
   B. 0
   C. -2
   D. 30

7. (1 point) Divide $(6x^4 + 7x^3 - 21x^2 + 9)$ by $(x - 3)$. What is the remainder?
   A. 0
   B. 9
   C. -6
   D. 12

8. (1 point) Factor $3x^4 - 8x^3 + 13x^2 - 30x$ completely. Which is NOT a factor?

A. x

B. $x - 2$

C. $3x - 5$

D. $x + 2$

9. (1 point) The polynomial $f(x) = x^5 + 2x^3 - x$ has how many x-intercepts?

A. 1

B. 2

C. 3

D. 5

10. (1 point) If $f(x) = x^4 - 4x^2$, which of the following is a turning point?

A. $(0, 0)$

B. $(2, 0)$

C. $(\sqrt{2}, -4)$

D. $(-2, 0)$

Long Answer (40 marks)

11. Polynomial Division

(a) (2 points) Divide $f(x) = x^3 - 4x^2 + 5x - 2$ by $x - 1$ and state the remainder.

(b) (2 points) Write the quotient and remainder in the form $f(x) = (x - 1)Q(x) + R$.

Solution: (a) Substitute $x = 1$:

$$f(1) = 1^3 - 4(1)^2 + 5(1) - 2 = 1 - 4 + 5 - 2 = 0$$

Remainder is 0.

(b) Perform polynomial division: Divide $x^3 - 4x^2 + 5x - 2$ by $x - 1$:

First term: $x^3/x = x^2$ **Multiply:** $x^2(x - 1) = x^3 - x^2$ **Subtract:** $(x^3 - 4x^2) - (x^3 - x^2) = -3x^2$ **Bring down** $+5x$**:** $-3x^2 + 5x$ **Next term:** $-3x^2/x = -3x$ **Multiply:** $-3x(x - 1) = -3x^2 + 3x$ **Subtract:** $(-3x^2 + 5x) - (-3x^2 + 3x) = 2x$ **Bring down** -2**:** $2x - 2$ **Next term:** $2x/x = 2$ **Multiply:** $2(x - 1) = 2x - 2$ **Subtract:** $(2x - 2) - (2x - 2) = 0$

So $Q(x) = x^2 - 3x + 2$, $R = 0$.

Final answer:

$$f(x) = (x - 1)(x^2 - 3x + 2)$$

12. Equation from Graph

(a) (2 points) A cubic function passes through $(0, 0)$, $(2, 0)$, and $(4, 0)$, and has a y-intercept of 0. Write its equation.

(b) (2 points) Sketch the graph, labeling all intercepts.

Solution: (a) Zeros at $x = 0, 2, 4$:

$$f(x) = ax(x-2)(x-4)$$

Since $f(0) = 0$, y-intercept is 0. a can be any nonzero constant.

(b) Sketch: - x-intercepts at $x = 0$, $x = 2$, $x = 4$ - y-intercept at $(0, 0)$ - For $a > 0$, as $x \to -\infty$, $f(x) \to -\infty$; as $x \to \infty$, $f(x) \to \infty$ - The graph crosses the x-axis at each root, with typical cubic shape: left end down, right end up, with two turning points between the roots.

13. Factoring Polynomials
    (a) (2 points) Factor $x^4 - 5x^3 + 6x^2$ completely.
    (b) (2 points) List all real zeros and their multiplicities.
    (c) (2 points) Sketch the graph, labeling all intercepts and describing the behavior at each zero.

Solution: (a) Factor out x^2:

$$x^4 - 5x^3 + 6x^2 = x^2(x^2 - 5x + 6) = x^2(x-2)(x-3)$$

(b) Real zeros: - $x = 0$ (multiplicity 2) - $x = 2$ (multiplicity 1) - $x = 3$ (multiplicity 1)

(c) Sketch: - x-intercepts at $x = 0$ (touches, does not cross), $x = 2$, $x = 3$ (crosses) - y-intercept at $f(0) = 0$ - Degree 4, positive leading coefficient: both ends up - At $x = 0$, graph touches and turns up (since multiplicity 2) - At $x = 2$ and $x = 3$, graph crosses the axis - The graph starts high on the left, touches at $x = 0$, dips down, crosses at $x = 2$, dips, crosses at $x = 3$, and rises to infinity on the right

14. Factoring with Rational Root Theorem
    (a) (2 points) Factor $2x^3 - 7x^2 + 6x$ completely.
    (b) (2 points) List all real zeros and their multiplicities.
    (c) (2 points) For each zero, describe how the graph behaves at that point.

Solution: (a) Factor out x:

$$2x^3 - 7x^2 + 6x = x(2x^2 - 7x + 6)$$

Factor quadratic:

$$2x^2 - 7x + 6 = (2x-3)(x-2)$$

So,

$$2x^3 - 7x^2 + 6x = x(2x-3)(x-2)$$

(b) Real zeros: - $x = 0$ (multiplicity 1) - $x = 1.5$ (multiplicity 1) - $x = 2$ (multiplicity 1)

(c) At each zero, the graph crosses the x-axis (all multiplicity 1).

15. Factoring Higher Degree
    (a) (2 points) Factor $x^5 - 3x^4 - 4x^3$ completely.

(b) (2 points) List all real zeros and their multiplicities.

(c) (2 points) Sketch the graph, labeling all intercepts and describing the behavior at each zero.

Solution: (a) Factor out x^3:

$$x^5 - 3x^4 - 4x^3 = x^3(x^2 - 3x - 4) = x^3(x-4)(x+1)$$

(b) Real zeros: - $x = 0$ (multiplicity 3) - $x = 4$ (multiplicity 1) - $x = -1$ (multiplicity 1)

(c) Sketch: - x-intercept at $x = 0$ (triple root: graph flattens and passes through) - $x = 4$ and $x = -1$ (crosses) - Degree 5, positive leading coefficient: left end down, right end up - At $x = 0$, graph flattens and passes through (triple root) - At $x = 4$ and $x = -1$, graph crosses the axis

16. Graph Analysis

(a) (2 points) Factor $f(x) = x^3 - 6x^2 + 11x - 6$ and list all zeros.

(b) (2 points) Find the coordinates of all turning points.

(c) (2 points) Sketch the graph, labeling all intercepts and turning points.

(d) (2 points) State the intervals where the function is increasing and decreasing.

Solution: (a) Factor:

$$x^3 - 6x^2 + 11x - 6 = (x-1)(x-2)(x-3)$$

Zeros: $x = 1, 2, 3$

(b) Find turning points by setting $f'(x) = 3x^2 - 12x + 11 = 0$:

$$x = \frac{12 \pm \sqrt{144 - 132}}{6} = \frac{12 \pm \sqrt{12}}{6} \approx 2.58, 1.42$$

Find y-values: At $x \approx 1.42$:

$$f(1.42) \approx (1.42)^3 - 6(1.42)^2 + 11(1.42) - 6 \approx 2.87 - 12.09 + 15.62 - 6 \approx 0.4$$

At $x \approx 2.58$:

$$f(2.58) \approx (2.58)^3 - 6(2.58)^2 + 11(2.58) - 6 \approx 17.18 - 39.89 + 28.38 - 6 \approx -0.33$$

Turning points: $(1.42, 0.4)$ **and** $(2.58, -0.33)$

(c) Sketch: - x-intercepts at $x = 1, 2, 3$ - y-intercept at $f(0) = -6$ - Turning points at $(1.42, 0.4)$ and $(2.58, -0.33)$ - Degree 3, positive leading coefficient: left end down, right end up

(d) Increasing: $x < 1.42$, $x > 2.58$ **Decreasing:** $1.42 < x < 2.58$

17. Graph from Factored Form

(a) (2 points) Given $f(x) = (x+2)^2(x-1)$, list all zeros and their multiplicities.

(b) (2 points) Describe the behavior of the graph at each zero.

(c) (2 points) State the end behavior.

(d) (2 points) Sketch the graph, labeling all intercepts and describing the behavior at each zero.

Solution: (a) Zeros: - $x = -2$ (multiplicity 2) - $x = 1$ (multiplicity 1)

(b) At $x = -2$, graph touches and turns (does not cross, since multiplicity 2). At $x = 1$, graph crosses the axis (multiplicity 1).

(c) Degree 3, positive leading coefficient: as $x \to -\infty$, $f(x) \to -\infty$; as $x \to \infty$, $f(x) \to \infty$.

(d) Sketch: - x-intercepts at $x = -2$ (touches), $x = 1$ (crosses) - y-intercept at $f(0) = (2)^2(-1) = 4(-1) = -4$ - Left end down, right end up - The graph touches at $x = -2$, crosses at $x = 1$, and passes through $y = -4$ at $x = 0$

18. Graph with Triple Root

(a) (2 points) Factor $f(x) = x^3 - 3x^2 + 3x - 1$ and list all zeros and their multiplicities.

(b) (2 points) Describe the behavior of the graph at the root.

(c) (2 points) Find the y-intercept.

(d) (2 points) Sketch the graph, labeling all intercepts and describing the behavior at the root.

Solution: (a) Factor:

$$x^3 - 3x^2 + 3x - 1 = (x - 1)^3$$

Zero: $x = 1$ (multiplicity 3)

(b) At $x = 1$, the graph has a point of inflection (triple root): it flattens as it passes through the axis.

(c) y-intercept: $f(0) = -1$

(d) Sketch: - x-intercept at $x = 1$ (triple root, point of inflection) - y-intercept at $f(0) = -1$ - Degree 3, positive leading coefficient: left end down, right end up - The graph flattens as it passes through $x = 1$